

Matplotlib

1. What is Matplotlib and why is it used?

Matplotlib is a widely used Python library for creating static, interactive, and animated visualizations from data. It provides flexible and customizable plotting functions that help in understanding data patterns, trends and relationships effectively.

Key Uses of Matplotlib:

- **Basic Plots:** Line plots, bar charts, histograms, scatter plots, etc.
- **Statistical Visualization:** Box plots, error bars, and density plots.
- **Customization:** Control over colors, labels, gridlines, and styles.
- **Subplots & Layouts:** Create multiple plots in a single figure.
- **3D Plotting:** Surface plots and 3D scatter plots using `mpl_toolkits.mplot3d`.
- **Animations & Interactive Plots:** Dynamic visualizations with `FuncAnimation`.
- **Integration:** Works well with Pandas, NumPy and Jupyter Notebooks.

2. Difference between line plot, bar plot, and scatter plot

Line Plot connect data points with lines across continuous intervals, such as time. They are ideal for displaying trends and patterns over time. Best for continuous data.

```
plt.plot(x, y)
```

Bar Plot use rectangular bars to represent numerical values across different categories. The length or height of each bar indicates the magnitude of the data. Bars can be vertical or horizontal.

```
plt.bar(x, y)
```

Scatter Plot represent relationships between two numerical variables using points on a 2D plane. Patterns or correlations can be visually identified.

```
plt.scatter(x, y)
```

3. What are figure and axes in Matplotlib?

- **Figure:** The overarching container that holds all plot elements, acting as the canvas for visualizations. It is the entire window or canvas where plots are drawn. It Can contain one or more Axes (plots). Controls overall properties like size, title, and layout
- **Axes:** The areas within the figure where data is plotted; each figure can contain multiple axes. Contains the x-axis, y-axis, labels, ticks, and plots. Multiple Axes can exist inside one Figure (subplots)

4. How do labels, titles, and legends improve data visualization?

Labels, titles, and legends make visualizations clear, informative, and easy to interpret.

1. Labels (X-label and Y-label): Explain what each axis represents

2. Titles: Give a clear summary of what the graph shows

3.Legends: Identify different data series or categories

5. Why is data visualization important before building AI models?

Data visualization helps understand data quality, patterns, and relationships, ensuring better preprocessing and more accurate AI models.

- **Understanding data distribution**
Visualization helps identify how data is spread (normal, skewed, imbalanced), which affects model selection and preprocessing.
- **Detecting outliers and anomalies**
Plots like box plots and scatter plots reveal outliers that can negatively impact AI model performance.

- **Identifying relationships between features**
Scatter plots and correlation heatmaps help understand feature relationships and their influence on the target variable.
- **Finding missing values and data issues**
Visual tools quickly highlight missing data patterns and inconsistencies that need cleaning before modeling.
- **Guiding feature selection and preprocessing**
Visualization helps decide which features are important, redundant, or need scaling or transformation.